CIS 215 LAB REPORT

Network Security Fundamentals

5.2.5 Lab — Configure Administrative Roles

BACKGROUND	
Course:	CIS 215 - Network Security Fundamentals
Term:	Spring 2026
Instructor:	Veliz E.
Section Number:	14381
Student Name:	Donaven Bruce
Student ID:	900*********
Submission Platform:	Canvas / File Upload as Assigned
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Week / Module Reference:	Week 5 / Router Administrative Access Control
Assignment Reference:	Lab 5.2.5 / Packet Tracer Activity
Lab Type:	Hands-On Lab / Packet Tracer
Date Completed:	2026-04-21
Time Spent:	Approximately 2.0 hours

ENVIRONMENT

Platforms, Devices, and Systems Used:

- Primary host system: Windows desktop environment with Cisco Packet Tracer used for build, configuration, testing, and evidence capture.
- Lab platform: Cisco Packet Tracer simulation with a three-router topology built manually from a blank workspace.
- Devices used: routers R1, R2, and R3; switches S1 and S3; hosts PC-A and PC-C.
- Router platform used in the simulation: ISR4321; switch platform used: 2960-24TT.
- Security focus area: role-based CLI administrative views with restricted command access on R1 and R3.

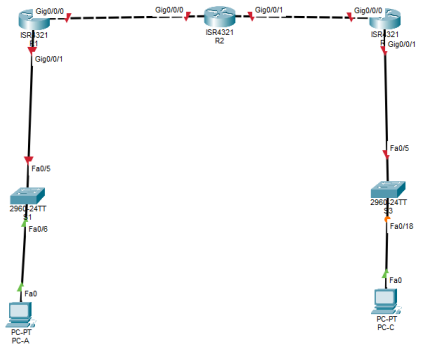

Figure 1. Initial Packet Tracer workspace after the required routers, switches, and PCs were placed and wired.

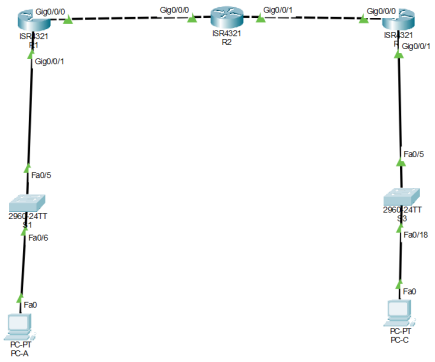

Figure 2. Operational topology after interfaces were brought up and the physical build was stable.

Files, Media, and Resources Used:

- Packet Tracer source file: completed 5.2.5 Configure Administrative Roles .pkt lab file.
- Lab instructions: assignment sheet containing the topology, addressing table, OSPF steps, and parser-view tasks.
- Evidence collected: initial build screenshots, routing verification screenshots, ping verification, tech-view restriction proof, and final parser-view / AAA / startup-config verification screenshots.
- Supporting notes: per-router interface map, addressing checklist, role passwords, and final save verification.

Work Objective:

- Build the required three-router network and cable every device to the interfaces specified in the addressing table.
- Configure interface addressing, host addressing, and OSPF routing, then verify full end-to-end communication.
- Create multiple administrative views on R1 and R3, grant different privilege levels, and verify restricted access behavior.

Learning Objectives:

- Configure router and host addressing from a supplied addressing table.
- Deploy OSPF process 1 and verify adjacency and learned routes across the routed path.
- Apply passive-interface correctly to the LAN-facing interfaces on R1 and R3.
- Enable AAA and configure parser views to enforce different levels of CLI access.
- Verify the difference between full-view, show-only, and limited-tech access through live command testing.

Scope and Constraints:

- In scope: topology build, interface bring-up, router addressing, host addressing, OSPF, connectivity verification, AAA, root view, parser views, command restriction validation, and final config save.
- The work was completed entirely inside Cisco Packet Tracer using manual device placement, manual cabling, and CLI-based configuration.
- The simulation used ISR4321 routers, which remained acceptable because the active interface identifiers matched the lab requirements and the intended outcomes were preserved.

Addressing Plan Used:

- R1 G0/0/0: 10.1.1.1 / 255.255.255.252
- R1 G0/0/1: 192.168.1.1 / 255.255.255.0
- R2 G0/0/0: 10.1.1.2 / 255.255.255.252
- R2 G0/0/1: 10.2.2.2 / 255.255.255.252
- R3 G0/0/0: 10.2.2.1 / 255.255.255.252
- R3 G0/0/1: 192.168.3.1 / 255.255.255.0
- PC-A NIC: 192.168.1.3 / 255.255.255.0 / default gateway 192.168.1.1
- PC-C NIC: 192.168.3.3 / 255.255.255.0 / default gateway 192.168.3.1

PART 1 — BASIC DEVICE SETTINGS

Preparation and Initial Router Configuration:

1) Placed and renamed R1, R2, R3, S1, S3, PC-A, and PC-C in the workspace.
2) Cabled R1 G0/0/0 to R2 G0/0/0, R2 G0/0/1 to R3 G0/0/0, R1 G0/0/1 to S1 F0/5, and R3 G0/0/1 to S3 F0/5.
3) Connected PC-A NIC to S1 F0/6 and PC-C NIC to S3 F0/18.
4) Entered privileged EXEC mode, set hostnames, and issued no shutdown on all active router interfaces.
5) Disabled DNS lookup on each router before beginning addressing and routing work.

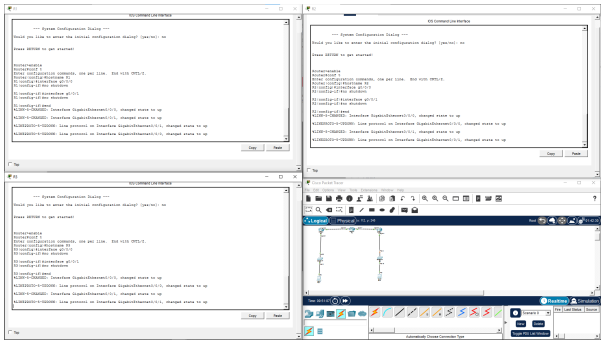

Figure 3. Initial router configuration stage after hostnames were applied and active interfaces were brought up.

Addressing and OSPF Procedure:

1) Applied the required interface IP addresses to R1, R2, and R3 exactly as listed in the lab addressing table.
2) Configured OSPF process 1 on all three routers.
3) Advertised 10.1.1.0/30 and 192.168.1.0/24 on R1.
4) Advertised 10.1.1.0/30 and 10.2.2.0/30 on R2.
5) Advertised 10.2.2.0/30 and 192.168.3.0/24 on R3.
6) Applied passive-interface G0/0/1 on R1 and R3 so the LANs were advertised without forming host-facing adjacencies.
7) Verified neighbor formation and route learning before moving to host validation.

```
R2>enable
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname R2
R2(config)#no ip domain-lookup
R2(config)#interface g0/0/0
R2(config-if)#ip address 10.1.1.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface g0/0/1
R2(config-if)#ip address 10.2.2.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#router ospf 1
R2(config-router)#network 10.1.1.0 0.0.0.3 area 0
R2(config-router)#network 10.2.2.0 0.0.0.3 area 0
R2(config-router)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

00:44:14: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0/0 from LOADING to FULL, Loading Done

00:44:21: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on GigabitEthernet0/0/1 from LOADING to FULL, Loading Done

R2#
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.1.1.2       YES manual up                    up
GigabitEthernet0/0/1   10.2.2.2       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.3.1       1   FULL/DR         00:00:38    10.2.2.1        GigabitEthernet0/0/1
192.168.1.1       1   FULL/DR         00:00:30    10.1.1.1        GigabitEthernet0/0/0
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.1.1.0/30 is directly connected, GigabitEthernet0/0/0
L       10.1.1.2/32 is directly connected, GigabitEthernet0/0/0
C       10.2.2.0/30 is directly connected, GigabitEthernet0/0/1
L       10.2.2.2/32 is directly connected, GigabitEthernet0/0/1
O    192.168.1.0/24 [110/2] via 10.1.1.1, 00:02:41, GigabitEthernet0/0/0
O    192.168.3.0/24 [110/2] via 10.2.2.1, 00:02:31, GigabitEthernet0/0/1

R2#
```

Figure 4. OSPF verification showing the required adjacencies in FULL state across the routed path.

```
R1>enable
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#interface g0/0/0
R1(config-if)#ip address 10.1.1.1 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface g0/0/1
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#router ospf 1
R1(config-router)#network 10.1.1.0 0.0.0.3 area 0
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#passive-interface g0/0/1
R1(config-router)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

00:44:18: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on GigabitEthernet0/0/0 from LOADING to FULL, Loading Done

R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.1.1.1       YES manual up                    up
GigabitEthernet0/0/1   192.168.1.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          1   FULL/BDR        00:00:33    10.1.1.2        GigabitEthernet0/0/0
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.1.1.0/30 is directly connected, GigabitEthernet0/0/0
L       10.1.1.1/32 is directly connected, GigabitEthernet0/0/0
O       10.2.2.0/30 [110/2] via 10.1.1.2, 00:01:50, GigabitEthernet0/0/0
   192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/1
O    192.168.3.0/24 [110/3] via 10.1.1.2, 00:01:50, GigabitEthernet0/0/0

R1#
```

Figure 5. R1 routing verification showing remote networks learned through OSPF.

Additional Notes for Part 1:

- The router interfaces changed to up/up after no shutdown was applied and the corresponding switch links finished initializing.
- The 10.1.1.0/30 and 10.2.2.0/30 transit links carried OSPF adjacency traffic, while the 192.168.1.0/24 and 192.168.3.0/24 LANs were advertised through passive interfaces.
- This staged workflow reduced troubleshooting time because physical state, addressing, and routing were validated in order.

PART 1 — CONTINUED

Host Configuration and Connectivity Validation:

1) Configured PC-A with IP 192.168.1.3 /24 and default gateway 192.168.1.1.
2) Configured PC-C with IP 192.168.3.3 /24 and default gateway 192.168.3.1.
3) Verified router-to-router reachability from R1 to the far-side R3 LAN interface.
4) Verified end-to-end host communication from PC-A to PC-C.
5) Observed one initial timeout due to ARP resolution, then repeated successful replies confirmed stable routed communication.
6) Saved the working Part 1 configuration before beginning administrative-role configuration.

```
R1#ping 192.168.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#
```

Figure 6. End-to-end host connectivity proof after routing converged and ARP resolution completed.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.3: bytes=32 time<1ms TTL=125
Reply from 192.168.3.3: bytes=32 time=1ms TTL=125
Reply from 192.168.3.3: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time<1ms TTL=253
Reply from 192.168.3.1: bytes=32 time<1ms TTL=253
Reply from 192.168.3.1: bytes=32 time<1ms TTL=253
Reply from 192.168.3.1: bytes=32 time<1ms TTL=253

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 7. Successful router-to-router verification from R1 to 192.168.3.1.

Key Part 1 Commands:

BASIC SETUP	ADDRESSING	ROUTING / VERIFY
enable	ip address 10.1.1.1 255.255.255.252	router ospf 1
configure terminal	ip address 10.1.1.2 255.255.255.252	network 10.1.1.0 0.0.0.3 area 0
hostname R1	ip address 10.2.2.1 255.255.255.252	network 10.2.2.0 0.0.0.3 area 0
hostname R2	ip address 10.2.2.2 255.255.255.252	network 192.168.1.0 0.0.0.255 area 0
hostname R3	ip address 192.168.1.1 255.255.255.0	network 192.168.3.0 0.0.0.255 area 0
no ip domain-lookup	ip address 192.168.3.1 255.255.255.0	passive-interface g0/0/1
interface g0/0/0	ping 192.168.1.1	show ip ospf neighbor
interface g0/0/1	ping 192.168.3.3	show ip route
no shutdown		ping 192.168.3.1
show ip interface brief		copy running-config startup-config
show running-config		

Part 1 Result:

- Part 1 was completed successfully because interfaces were active, OSPF adjacencies formed, remote routes were learned, and both router-to-router and host-to-host communication were verified.

PART 2 — ADMINISTRATIVE ROLES

Administrative Role Configuration Procedure:

1) Enabled AAA on R1 and R3 with aaa new-model.
2) Configured enable secret cisco12345 to support root-view access.
3) Entered root view with enable view and created admin1, admin2, and tech on R1.
4) Configured admin1 for show, configure terminal, and debug access.
5) Configured admin2 for show-only access.
6) Configured tech with only show version, show interfaces, show ip interface brief, and show parser view.
7) Repeated the same role configuration sequence on R3.

Role View Verification and Contrast:

- admin1 was verified as the highest non-root view and allowed configure and debug commands in addition to show.
- admin2 was verified as a junior-view role and did not include configure or debug.
- tech was verified as a restricted operational role with only selected show commands.
- show ip interface brief succeeded in tech view because that specific command was explicitly added.
- show ip route failed in tech view because it was not included in the tech view command set.
- IOS also displayed parent commands such as show, show ip, and show ip interface because those are required levels in the command hierarchy.

```
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#enable view tech
Password:
R1#%PARSER-6-VIEW_SWITCH: successfully set to view 'tech'.


R1#show parser view
Current view is 'tech'
R1#?
Exec commands:
  disable     Turn off privileged commands
  enable      Turn on privileged commands
  exit        Exit from the EXEC
  logout      Exit from the EXEC
  show        Show running system information
R1#
R1#show ?
  interfaces          Interface status and configuration
  ip                  IP information
  parser              Show parser commands
  version             System hardware and software status
R1#show
% Incomplete command.
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.1.1.1        YES manual up                    up
GigabitEthernet0/0/1   192.168.1.1     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#show ip route
                 ^
% Invalid input detected at '^' marker.

R1#
```

Figure 8. Tech-view restriction proof showing that show ip interface brief worked while show ip route was denied.

```
R1#
R1#show run | section parser view
parser view admin1
 secret 5 $1$mERr$ILrAmVhMGbrCFnj8QqS3T.
 commands exec include configure
 commands exec include all configure terminal
 commands exec include all debug
 commands exec include all show
parser view admin2
 secret 5 $1$mERr$Y86euNWSPLMQo.DAj6vGJ0
 commands exec include all show
parser view tech
 secret 5 $1$mERr$QMUdu4f7Qgk/Gy0quE8Eh0
 commands exec include show
 commands exec include show interfaces
 commands exec include show ip
 commands exec include show ip interface
 commands exec include show ip interface brief
 commands exec include show parser
 commands exec include show parser view
 commands exec include show version
R1#show run | include aaa|enable secret
enable secret 5 $1$mERr$WvpW0n5HghRrqnrwXCUU1.
aaa new-model
R1#show startup-config | section parser view
                       ^
% Invalid input detected at '^' marker.

R1#
```

Figure 9. Final configuration proof showing parser views, AAA, and saved startup configuration evidence.

Key Part 2 Commands:

ROOT / AAA	VIEWS	TECH / VERIFY
aaa new-model	parser view admin1	parser view tech
enable secret cisco12345	secret admin1pass	secret techpasswd
enable view	commands exec include all show	commands exec include show version
show run | include aaa|enable secret	commands exec include all config terminal	commands exec include show interfaces
copy running-config startup-config	commands exec include all debug	commands exec include show ip interface brief
	parser view admin2	commands exec include show parser view
	secret admin2pass	show parser view
	commands exec include all show	show ip interface brief
		show ip route
		show run | section parser view
		show startup-config

Questions Answered During Verification:

1) What is missing from admin2 that is present in admin1? - configure and debug.
2) Were you able to do show ip interface brief as the tech user? - Yes, because it was explicitly added.
3) Were you able to do show ip route as the tech user? - No, because it was not included.
4) Why are show, show ip, show ip interface, and show ip interface brief all listed? - IOS includes the parent command hierarchy required to reach the full command.

Part 2 Result:

- Part 2 was completed successfully because the three views were created on both R1 and R3, their privileges behaved as expected, and the final configurations were written to startup-config.

SUMMARY, RESULTS, AND REFLECTION

Completion Verification:

- The topology was built manually and all required router, switch, and host connections were completed correctly.
- Router addressing, PC addressing, OSPF routing, passive interfaces, and end-to-end host communication were verified.
- Administrative roles admin1, admin2, and tech were created and verified on both R1 and R3.
- AAA and enable secret were confirmed in the running configuration and the startup configuration on both routers.
- The final Packet Tracer file was saved in a working, submission-ready state after all validation steps were complete.

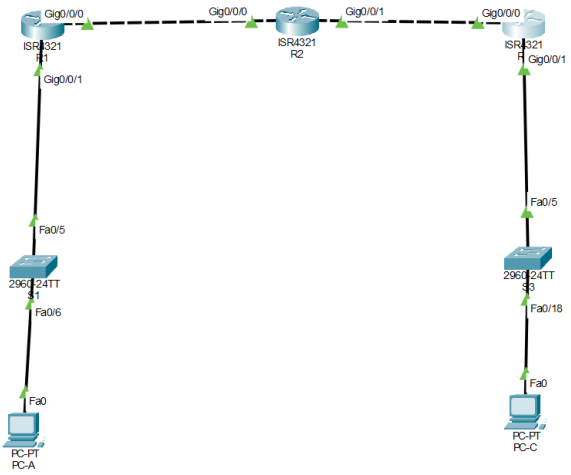

Figure 10. Final completed topology after routing and administrative-role verification were fully complete.

Key Findings:

1) A correct routed baseline is essential before administrative security features can be verified confidently.
2) Role-based CLI views provide meaningful separation between full administrative access, show-only access, and limited technician access.
3) Verification must include both allowed and denied commands, because configuration text alone does not prove access control behavior.
4) Saving and re-checking startup-config is valuable because it confirms the roles persist beyond the live running session.

Security Concepts Reinforced:

- Administrative access is part of the attack surface, so limiting commands by role reduces unnecessary exposure.
- AAA and root-view authentication control who can create or modify privileged administrative views.
- Least privilege is practical: a technician may need interface visibility without route visibility or configuration authority.
- Verification should test successful commands and denied commands so access boundaries are proven operationally.

Best Practices / Operational Notes:

- Complete the physical topology and routing baseline first, then layer on access control and administrative restrictions.
- Test both what a role can do and what a role cannot do, since denial behavior is part of the security objective.
- Save and inspect startup-config at the end so the result is not only working live but also persistent for later validation.

Reflection:

- This lab reinforced that router security is not only about protecting packet flow but also about controlling who can do what at the CLI.
- The most useful takeaway was seeing the difference between admin1, admin2, and tech in live command behavior rather than only reading the view definitions in the running configuration.
- From a professional standpoint, the completed work demonstrates the ability to build, route, verify, restrict, and persist a small multi-router network using a clean phased workflow.

Artifacts and Deliverables:

- Completed Packet Tracer .pkt file: 5.2.5-configure-administrative-roles.pkt
- Supporting screenshots for initial build, routing verification, connectivity validation, restriction proof, and saved-config proof.
- Final written report documenting the completed lab workflow, administrative-role configuration, and verification results.

Next Steps:

- Attach the completed Packet Tracer file and submit it through Canvas as required by the assignment.
- Store the final screenshots and report in the course repository as polished evidence of the completed lab.
- Reuse the same phased workflow in future labs involving AAA, ACLs, privilege separation, and broader management-plane security.